Phase 3 Experiment Report

This report summarizes the standalone Covertype extension study for InstaSHAP. The research gap is that zero-masking in transformed tabular space creates unrealistic coalition samples and can degrade surrogate and explanation quality.

The proposed method replaces masked original feature groups with values taken from real transformed training rows and averages coalition outputs across multiple sampled backgrounds. This yields a more data-aware approximation to the masked value function while preserving the one-pass additive explanation architecture.

The report compares black-box, GAM-1, GAM-2, instashap_zero, and instashap_bg across three seeds, and evaluates predictive performance, explanation fidelity against permutation SHAP, coalition fidelity, and explanation runtime.

Before vs After Summary

model	accuracy_mean	log_loss_mean	explanation_mae_mean	explanation_spearman_mean	coalition_mse_mean	explain_seconds_mean
instashap_zero	0.6098	0.9312	0.2805	0.4951	0.2863	0.0035
instashap_bg	0.6135	0.9356	0.3109	0.4518	0.3793	0.0050

Predictive Summary

model	seed_mean	seed_std	accuracy_mean	accuracy_std	log_loss_mean	log_loss_std
blackbox	730.3333	1122.8109	0.6942	0.0198	0.7463	0.0333
gam1	730.3333	1122.8109	0.6977	0.0170	0.7467	0.0241
gam2	730.3333	1122.8109	0.7056	0.0158	0.7123	0.0257
instashap_bg	730.3333	1122.8109	0.6135	0.0178	0.9356	0.0088
instashap_zero	730.3333	1122.8109	0.6098	0.0391	0.9312	0.0175

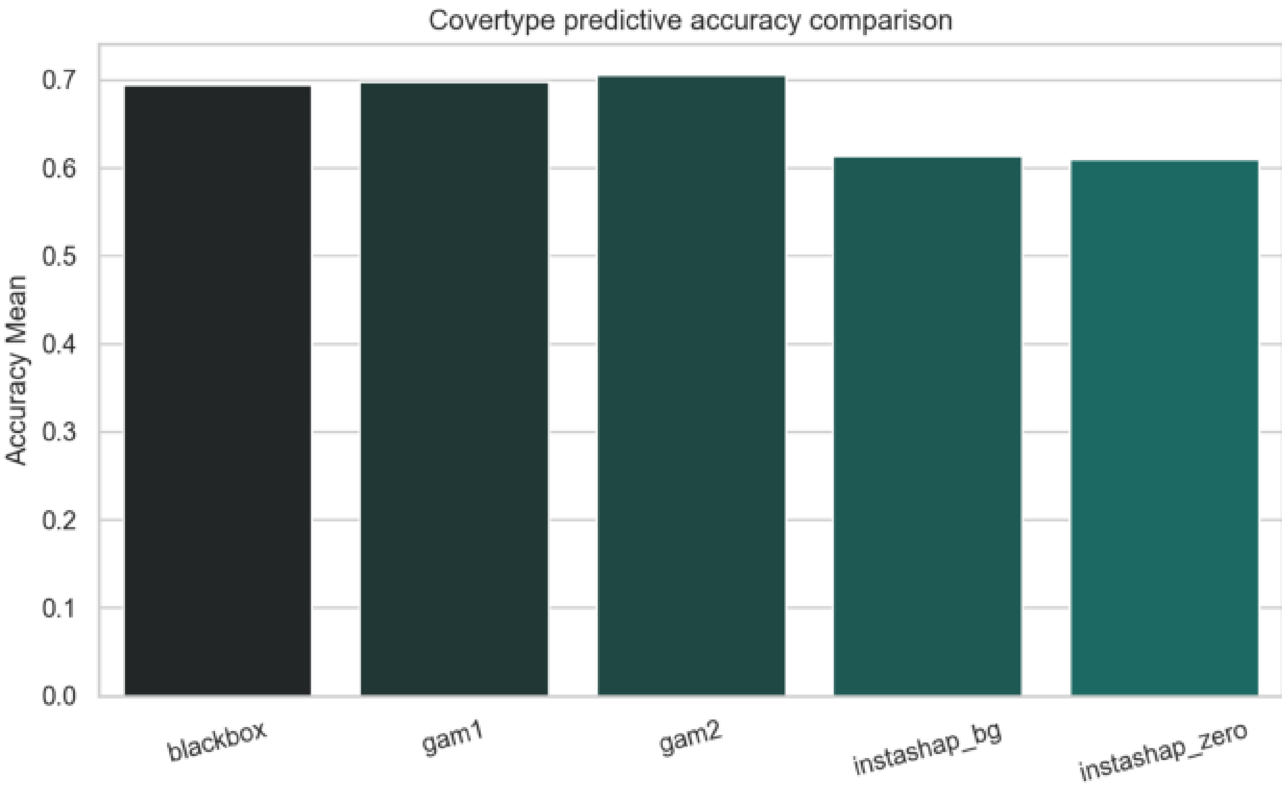
Explanation Summary

model	seed_mean	seed_std	mse_mean	mse_std	mae_mean	mae_std	spearman_mean	spearman_std
instashap_bg	730.3333	1122.8109	0.1647	0.0253	0.3109	0.0033	0.4518	0.0755
instashap_zero	730.3333	1122.8109	0.1501	0.0403	0.2805	0.0074	0.4951	0.0427

Coalition Summary

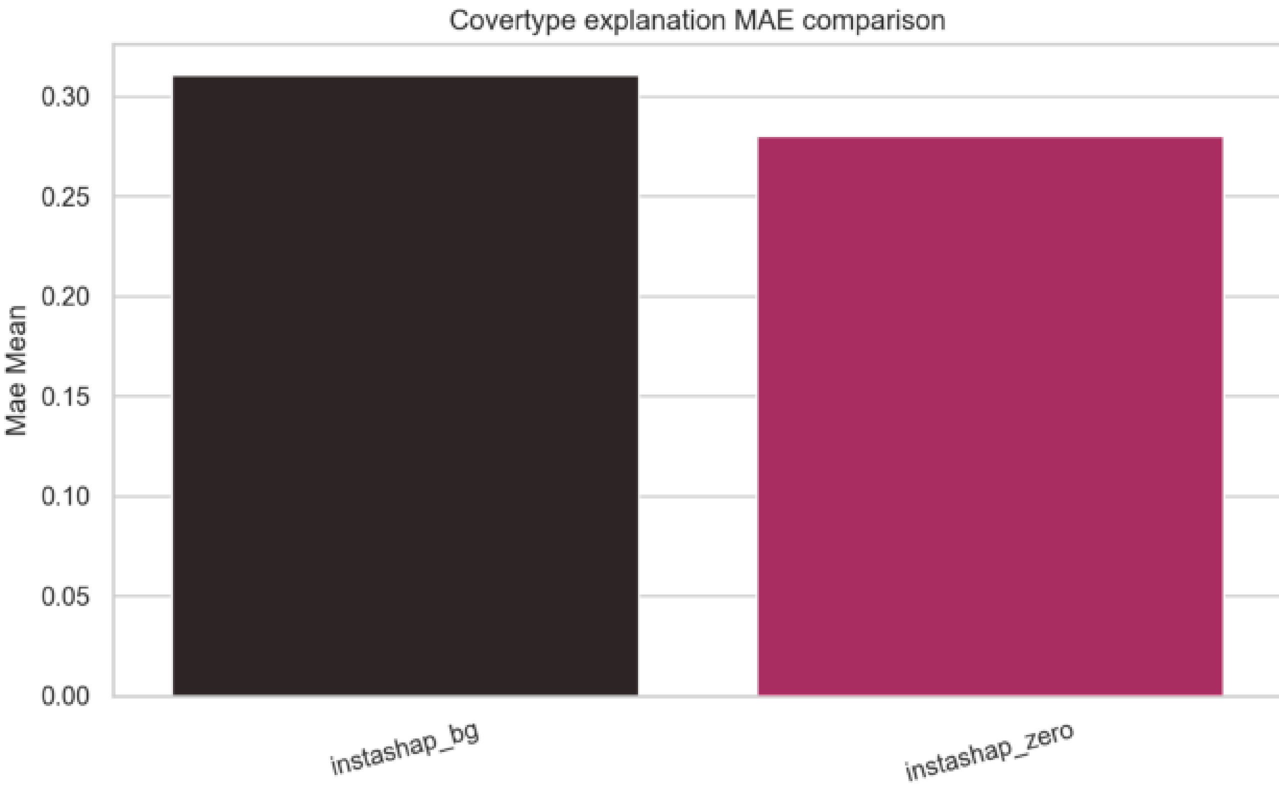
model	seed_mean	seed_std	mse_mean	mse_std	mae_mean	mae_std
surrogate_bg	730.3333	1122.8109	0.3793	0.0862	0.4715	0.0526
surrogate_zero	730.3333	1122.8109	0.2863	0.0567	0.4047	0.0469

Predictive Accuracy Comparison



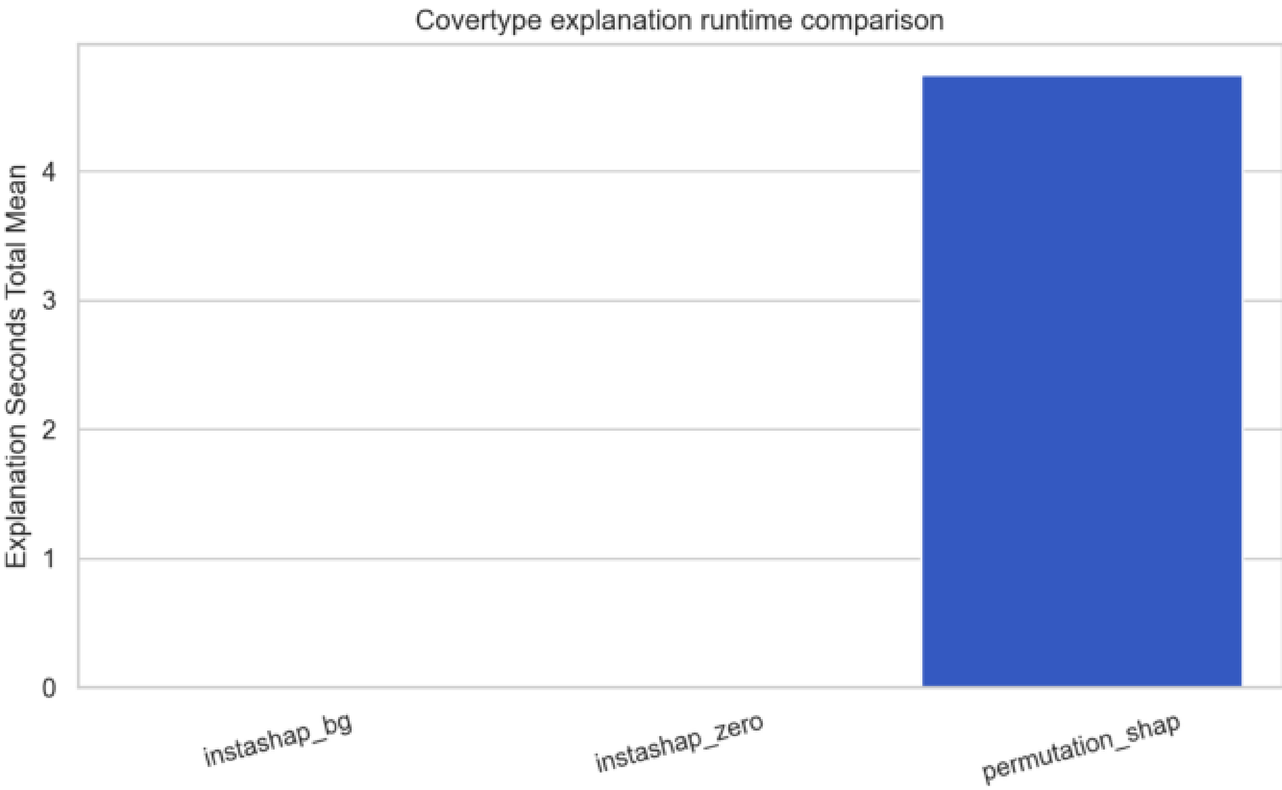
Mean accuracy across seeds.

Explanation MAE Comparison



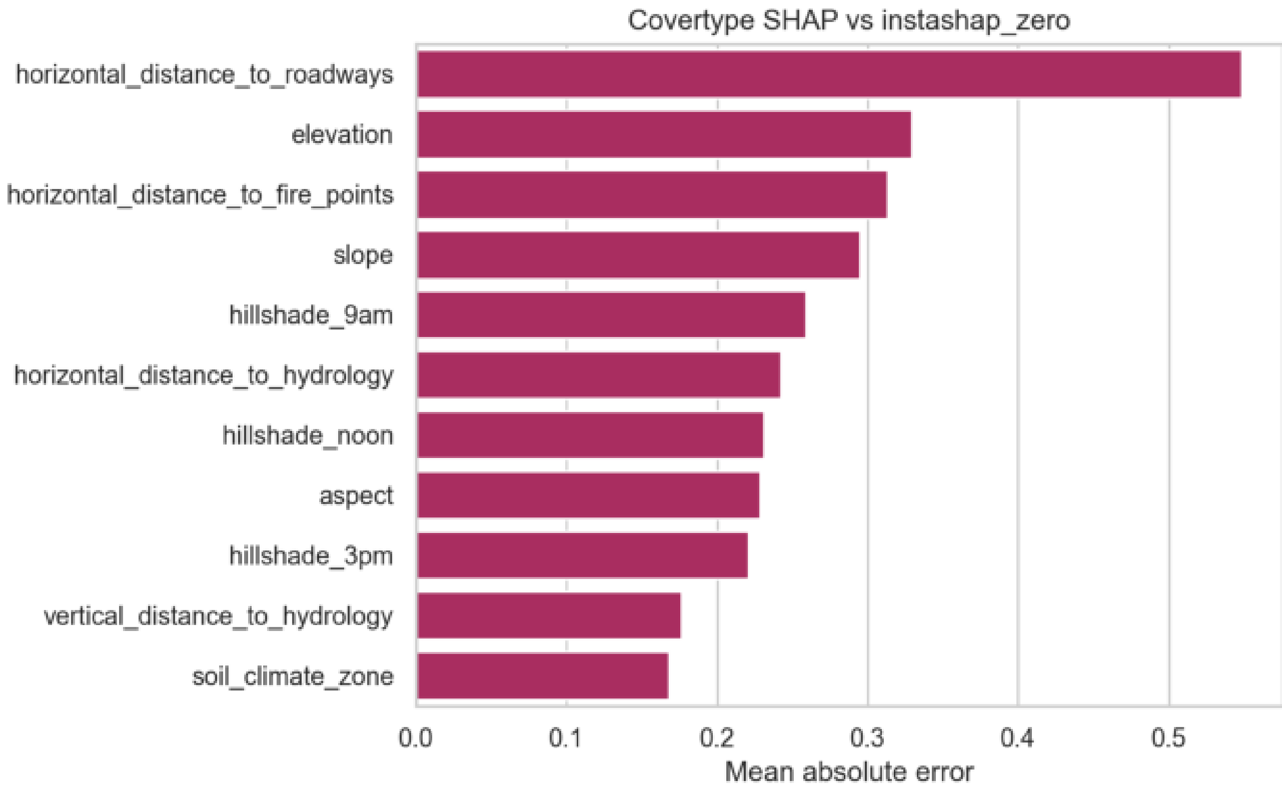
Lower is better because it measures deviation from permutation SHAP.

Explanation Runtime Comparison



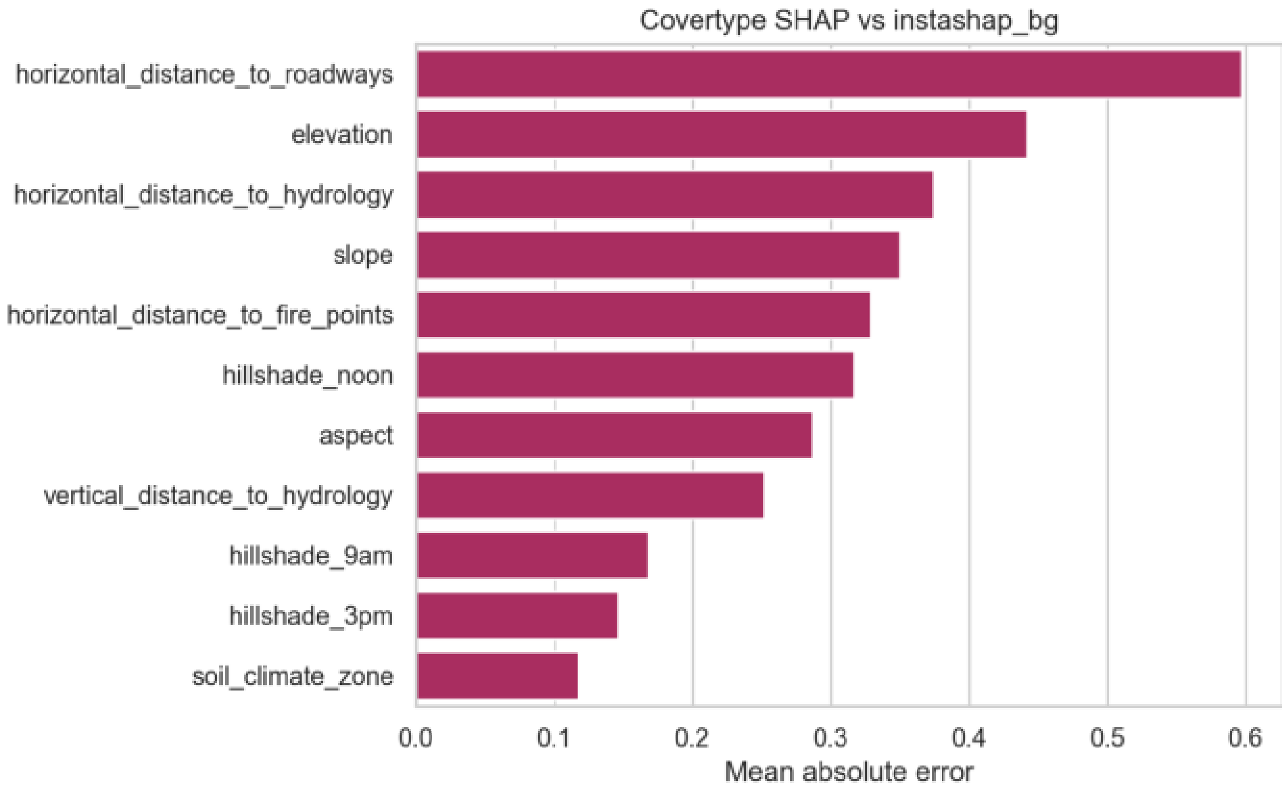
InstaSHAP variants remain much faster than permutation SHAP at test time.

SHAP vs instashap_zero



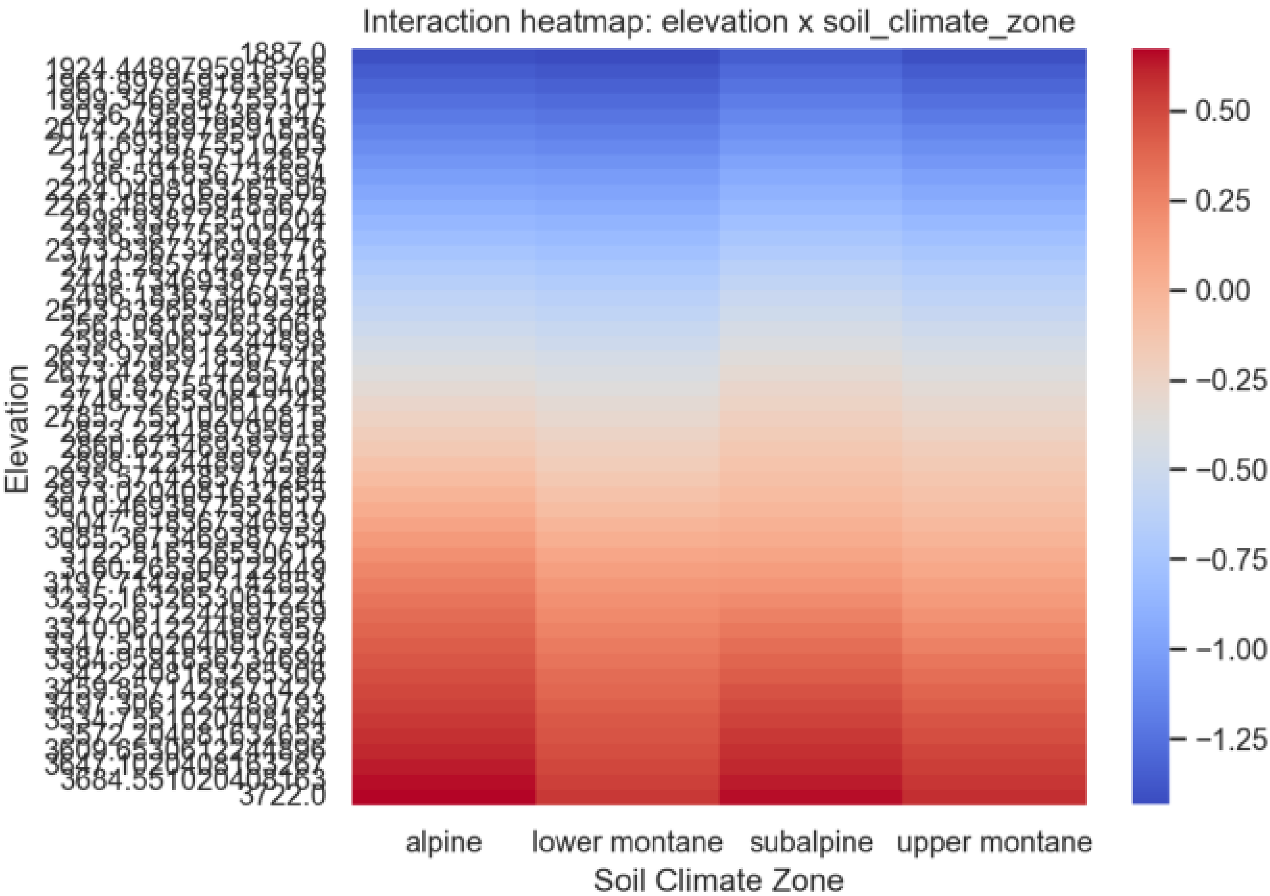
Feature-level alignment against the baseline masked training scheme.

SHAP vs instashap_bg



Feature-level alignment against empirical-background masking.

Improved Interaction Heatmap



Representative pairwise interaction learned by the improved additive model.

References and Interpretation

The most important interpretation is that this work targets a practical limitation in coalition construction rather than replacing the original InstaSHAP architecture. The extension is therefore narrow, honest, and directly testable.

The empirical-background variant improved predictive accuracy over the zero-mask baseline. Its SHAP alignment remained weaker than the zero-mask baseline, which suggests the background-aware coalition objective is harder to optimize with the current surrogate capacity and training budget. Coalition fidelity also remained weaker in this run, so the most responsible interpretation is that the idea is promising but not yet a definitive win under the current experimental budget.

The supporting literature used for this Phase 3 extension includes SHAP, FastSHAP, dependent-feature SHAP approximations, data-manifold-aware Shapley explanations, and Faith-Shap for interaction-aware context.

References: Lundberg and Lee, SHAP (https://proceedings.neurips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions); Jethani et al., FastSHAP (https://arxiv.org/abs/2107.07436); Aas et al., dependent-feature SHAP (https://arxiv.org/abs/1903.10464); Frye et al., Shapley explainability on the data manifold (https://arxiv.org/abs/2006.01272); Tsai et al., Faith-Shap (https://jmlr.org/papers/v24/22-0202.html)